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MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143 ELECTRICAL PARTS AND ASSEMBLIES FOR

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, tests, manufacture and acceptance requirements for Electrical Parts, Fire Board Assemblies and Logic Board Assemblies and Electrical Assemblies (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-27	- Transformer and Inductor, General Specification for
MIL-S-19500	- Semiconductor Devices, General Specification for
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-39006/22 - Capacitors, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85°C (Voltage Derated to 125°C), Established Reliability, Style CLR79

MIL-S-63257 - Safety and Arming Device, GM: M143 Loading, Assembling and Packing For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts

MIL-STD-750 - Test Methods for Semiconductor Devices

MIL-STD-1285 - Marking of Electrical and Electronic Parts

MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

9311342 - Fire Board Assembly
9311372 - Logic Circuit Board Assembly
9321558 - Connector Cable Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Material. Materials, and parts components shall be in accordance with applicable drawings and specifications.

3.2 Assemblies. The complete assemblies shall comply with all requirements specified on Drawing (Dwg.) 9311342, 9311372 and 9321558 and associated drawings and with all requirements specified in the applicable specifications and standards.

3.3 Fire board assembly (Dwg. 9311342).

3.3.1 Continuity. The electrical resistance between the following printed board terminals shall be no greater than 0.2 ohms:

a. E11 to E15
b. E12 to E14
c. E13 to E16

3.3.2 Resistance. The electrical resistance between the ground terminal E1 (except k) to the terminals below shall be as specified in Table I while applying an open electrical potential no greater than 10VDC.

TABLE I - Resistance

IDENTIFICATION	TERMINAL	REQUIREMENT (OHMS)
a. Fire Signal Input	E2	46 to 68
b. Detonator Input	E5	1.6 MEG to 2.4 MEG
c. Destruct Input	E6	1.6 MEG to 2.4 MEG
d. Fire TM Input	E7	Greater than 20 MEG
e. Power Input	E8	8.4K to 12.6K
f. ITL Command	E9	Greater than 20 MEG
g. ITL Command Return	E10	Greater than 20 MEG
h. ITL TM	E11	Greater than 20 MEG
i. ITL Monitor	E12	Greater than 20 MEG
j. ITL Monitor	E13	Greater than 20 MEG
k. Resistor, Diode	(+)E3 to (-)E4	2.4K to 19 MEG

3.3.3 Fire circuit characteristics.

3.3.3.1 Fire circuit charging. The fire circuit capacitor voltage shall be 70.0VDC minimum (min.) at (+)E4 to (-)E1 at a maximum (max.) of 2.50 seconds after 21.5 - 0.5VDC is applied to (+)E8 to (-)E1. The initial voltage on (+)E4 to (-)E1 shall be less than 1.5VDC prior to start of test.

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3.3.3.2 Fire circuit capacitor voltage storage. The fire circuit capacitor voltage shall remain 70.0VDC min. at (+)E4 to (-)E1 six (6) seconds min. after the 21.5 - 0.5VDC charging voltage is removed from (+)E8 to (-)E1.

3.3.3.3 No charging. With a 100 ± 10%, 1/4 watt min. resistor connected between E5 and E1, the fire circuit capacitor voltage (+)E4 to (-)E1 shall be no greater than 15VDC after applying an 18 ± 2.0V pulse with a duration of 500 ± 100msec at (+)E2 to (-)E1. Within 100 + 10msec of the application of the 18 ± 2.0V pulse apply 28 ± 2.8VDC at (+)E8 to (-)E1 for a period not to exceed 10 - 5 sec.

3.3.3.4 Fire circuit stabilized voltage. The fire circuit capacitor voltage shall be 70.0VDC min. to 105.0VDC max. at (+)E4 to (-)E1 after a min. of 10 seconds charging with 21.5 - 0.5VDC applied at (+)E8 to (-)E1.

3.3.4 Fire circuit function. With a 100 ohms ± 10%, 1/4 watt min. resistor connected between E5 and E1, the fire circuit is defined as having functioned when the fire circuit capacitor voltage (+)E4 to (-)E1 has discharged from 70.0VDC min. to less than 15.0VDC in no greater than 400msec after a fuze fire signal or logic destruct signal is applied.

3.3.4.1 Logic destruct. The fire circuit shall function when 1.0 ± 0.2VDC is applied at (+)E6 to (-)E1 for 100 ± 20msec.

3.3.4.2 No fire signal A. The fire circuit shall not function when a 5.0 ± 0.25V pulse for a duration of 200 ± 10 microseconds with a rise time no greater than one (1) microsecond is applied at (+)E2 to (-)E1.

3.3.4.3 No fire signal B. The fire circuit shall not function when a 10.0 ± 0.5V pulse for a duration of 2.0 + 0.5 microseconds with a rise time no greater than one (1) microsecond is applied at (+)E2 to (-)E1.

3.3.4.4 No fire signal C. The fire circuit shall not function when a 30.0 ± 1.5V pulse for a duration of 0.4 ± 0.05 microseconds with a rise time no greater than 0.08 microseconds is applied at (+)E2 to (-)E1.

3.3.4.5 Fuze fire signal. The fire circuit shall function when a fuze fire signal pulse of 18.0 - 2.0V for a duration of 90 - 10 microseconds with a rise time no greater than 20 microseconds is applied at (+)E2 to (-)E1.

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3.4 Logic circuit board assembly (Dwg. 9311372). All measurements required in this specification shall be taken with a measurement device that has an input impedance equal to or greater than 10 megohms and an open circuit potential no greater than 10.0VDC. Prior to start and for duration of testing, the following load and biasing resistors shall be added externally to the logic circuit board under test.

PIN (+) TO (-)	RESISTOR VALUE (OHMS)
E26 - E3	100K
E25 - E3	100K
E23 - E3	1K
E22 - E3	1K (2 watt min.)
E1 - E3	47K
E2 - E3	47K
E6 - E3	47K
E24 - E3	1K

Unless otherwise specified, all resistor are 1/4 watt min., ± 10%.

3.4.1 Continuity and resistance. The electrical resistance shall be as specified in Table II.

TABLE II - Continuity and Resistance

TERMINAL (+)	TERMINAL (-)	REQUIREMENTS (OHMS)	TERMINAL (+)	TERMINAL (-)	REQUIREMENTS (OHMS)
E4	E3	0.3 max.	E13	E12	120K to 240K
E29	E3	90 to 110	E19	E3	0.9 to 1.1 MEG
E30	E3	90 to 110	E18	E3	0.9 to 1.1 MEG
E31	E15	90 to 110	E15	E3	0.9 to 1.1 MEG
E20	E21	Less than 19 MEG	E16	E32	0.99 to 1.01 MEG
E28	E3	80K to 120K	E17	E33	0.99 to 1.01 MEG
E27	E12	120K to 240K	E21	E20	Greater than 20 MEG
E27	E13	120K to 240K	E5	E3	Greater than 20 MEG
E12	E13	120K to 240K			

3.4.2 Charge command test.

3.4.2.1 TM monitors. With 100 + 5.0VDC applied at right Convertor input (+)E25 to (-)E3 and followed by 31.0 - 0.5VDC applied at Charge Command input (+)E2 to (-)E3, after 0.5sec min. the voltage between the indicated terminals to (-)E3 shall be as specified in Table III.

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TABLE III - TM monitors

	IDENTIFICATION	TERMINAL (+)	REQUIREMENT (volts)
a.	Power/Destruct	E1	-0.5 to +0.5
b.	5VDC Test	E6	-0.5 to +0.5
c.	ITL TM	E8	3.5 to 5.5
d.	Fire TM	E9	-0.5 to +0.5
e.	Arm TM	E10	3.5 to 5.5
f.	Rotor Monitor, Right	E12	3.5 to 5.5
g.	Rotor Monitor, Left	E13	3.5 to 5.5
h.	5VDC (Regulator) Monitor	E14	3.5 to 5.5
i.	Power Output (Diode Drop)	E22	28.5 to 31.0
j.	Convertor Input, Left	E26	-0.5 to +0.5
k.	ITL Unlock	E27	3.5 to 5.5
l.	5VDC	C1(+)	3.5 to 5.5
m.	5VDC	C2(+)	3.5 to 5.5

3.4.2.2 Fire TM. With the voltages applied as specified in 3.4.2.1, the Fire TM voltage at (+)E9 to (-)E3 shall be 4.5 ± 1.0VDC when 5.0 ± 0.5VDC is applied at (+)E28 to (-)E3.

3.4.2.3 Arm TM. With the voltages applied as specified in 3.4.2.1, the Arm TM voltage at (+)E10 to (-)E3 shall be 0 ± 0.5VDC when E12 is connected to E3. With E12 removed from E3 and E13 connected to E3, the Arm TM shall be 0 ± 0.5VDC.

3.4.2.4 ITL TM. With the voltages applied as specified in 3.4.2.1, the ITL TM voltage at (+)E8 to (-)E3 shall be 0 ± 0.5VDC when a 100 ohm ± 10 percent resistor is connected between E27 and E3 ground. Remove the 100 ohm resistor at completion of measurement.

3.4.2.5 Convertor TM. Apply the Charge Command voltage and remove the right convertor voltage as specified in 3.5.2.1. After 0.5sec min. the voltage at (+)E11 to (-)E3 shall be 0 ± 0.5VDC and the Destruct Voltage (+)E23 right and (+)E24 left to (-)E3 shall be 1.5VDC min.

3.4.2.6 Destruct pulse.

3.4.2.6.1 Power up. The voltage on both 68 microfarad capacitors shall be less than 0.5VDC prior to the start of this test. After the voltage is applied as specified in 3.5.2.1, the destruct pulse at (+)E23 right and (+)E24 left to (-)E3 shall change from 0 ± 0.5VDC to 1.5VDC min. in less than 50msec. The duration of the Destruct pulse shall be not less than 2 msec and not greater than 200msec.

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3.4.2.6.2 Convertor input removed. With the voltage applied as specified in 3.5.2.1, for 0.5sec min., the Destruct pulse at (+)E23 right and (+)E24 left to (-)E3 shall be 1.5VDC min. in not less than 10msec and not greater than 50msec after the right convertor voltage is removed.

3.4.2.7 Capacitor storage. Apply 5.0 + 1.0VDC at 5VDC Test (+)E6 to (-)E3, 100 + 5.0VDC at converter input (+)E25 to (-)E3, and 31 - 0.5VDC at Charge Command (+) E2 to (-)E3 and Logic Board Power output (+)E22 to (-)E3. These voltages shall be applied for not less than two (2) seconds prior to this test. Remove the voltage at (+)E2 to (-)E3 and after a 10 ± 1 second interval remove the voltages at (+)E6 to (-)E3 and (+) E22 to (-)E3. The output (Destruct Pulse) at (+)E23, Right and (+)E24, Left to (-)E3 shall be 1.0VDC min. for 50 milliseconds min. upon removal of voltage at (+)E6 to (-)E1.

3.4.3 Power/destruct tests.

3.4.3.1 TM monitors. With 100 + 5.0VDC applied to Left Convertor (+)E26 to (-)E3 and 23.0 + 0.5VDC applied to Power/Destruct (+)E1 to (-) E3, the voltage across the terminals to (-)E3 shall be as specified in Table IV.

TABLE IV - TM monitors

	IDENTIFICATION	TERMINAL (+)	REQUIREMENT (volts)
a.	Charge Command	E2	-0.5 to +0.5
b.	5VDC Test	E6	-0.5 to +0.5
c.	ITL TM	E8	3.5 to 5.5
d.	Fire TM	E9	-0.5 to +0.5
e.	Arm TM	E10	3.5 to 5.5
f.	Rotor Monitor, Right	E12	3.5 to 5.5
g.	Rotor Monitor, Left	E13	3.5 to 5.5
h.	5VDC (Regulator) Monitor	E14	3.5 to 5.5
i.	Power Output (Diode Drop)	E22	21.0 to 22.5
j.	Converter Input, Right	E25	-0.5 to +0.5
k.	ITL TM	E27	3.5 to 5.5
l.	5VDC	C1(+)	3.5 to 5.5
m.	5VDC	C2(+)	3.5 to 5.5

3.4.3.2 Power TM threshold. With 28 ± 3VDC initially applied to Power/Destruct (+)E1 to (-)E3 and the left convertor voltage applied as specified in 3.4.3.1, the Power TM voltage at (+)E7 to (-)E3 shall be 4.5 ±1.0VDC, when the Power/Destruct input applied at (+)E1 to (-)E3 is equal to or greater than 25.0VDC. The Power TM voltage at (+)E7 to (-)E3 shall be 0 ± 0.5VDC, when the Power/Destruct input applied to (+)E1 to (-)E3 is less than 21.0VDC.

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3.4.3.3 Destruct pulse threshold. With the voltages initially applied as specified in 3.4.3.1, the Destruct voltage at both terminals (+)E23 right and (+)E24 left to (-)E3 shall be 0 ± 0.5VDC when the Power/Destruct input applied at (+)E1 to (-)E3 is equal to or greater than 20.5VDC. The Destruct voltage at both (+)E23 right and (+)E24 left to (-)E3 shall be 1.5VDC min. when the Power/Destruct input applied at (+)E1 to (-)E3 is less than 18.5VDC.

3.4.3.4 Converter threshold. With the voltages initially applied as specified in 3.4.3.1, the Destruct voltage at both (+)E23 right and (+)E24 left to (-)E3 shall be 0 ± 0.5VDC and the convertor TM at (+)E11 to (-)E3 shall be 4.5 ± 1.0VDC when the left convertor voltage (+)E26 to (-)E3 is greater than 70.0VDC. The destruct voltage at (+)E23 right and (+)E24 left to (-)E3 shall be 1.5VDC min. and the Convertor TM at (+)E11 to (-)E3 shall be 0 ± 0.5VDC when the voltage on (+)E26 to (-)E3 is less than 60.0VDC. This transition shall occur at 65.0 ± 5VDC.

3.4.4 Five-volt test. Apply 100 + 5.0VDC to left convertor (+)E26 to (-)E3, 23.0 + 0.5VDC to Power/Destruct (+)E1 to (-)E3, 23.0 + 0.5VDC to logic board power output (+)E22 to (-)E3 and 5.0 ± 0.5VDC to 5VDC test (+)E6 to (-)E3 for the following test.

3.4.4.1 Destruct pulse. Wait a min. of 0.5 seconds then remove the 23VDC at E1 followed by removing the 5VDC at E6 as specified in 3.4.4. The Destruct voltage at (+)E23 right and (+)E24 left to (-)E3 shall change from 0 ± 0.5VDC to 1.5VDC min. in not less than 10msec and not greater than 50msec.

3.5 Connector cable assembly - (Dwg 9321558).

3.5.1 Continuity and isolation. When each of the points listed below is activated with a logic level high, the corresponding points indicated shall be at a logic level high and all others shall be at a logic level low. Abbreviations are as follows: J1 = connector pin; L = logic board connection; FR = right fire board connection; FL = left fire board connection; SR = right rotor switch connection; SL = left rotor switch connection.

Parts Activated	Points Requiring Logic High
J1-1	L6
J1-2	L13, SL1
J1-3	FR13, L18
J1-4	L8
J1-5	FL13, L19
J1-6	J1-34

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Parts Activated	Points Requiring Logic High
J1-7	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-8	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2,
J1-9	All Low
J1-10	All Low
J1-11	All Low
J1-12	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2,
J1-13	J1-45, J1-40, J1-39, J1-29, J1-24, J17, J1-16, J1-15, J1-14, J1-12, J1-8, J1-7,L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2,
J1-14	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-15	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-16	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-115, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-17	J1-45, J1-40, J1-39, J1-29, J1-24, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7,L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-18	L12, SR1
J1-19	L10
J1-20	L9
J1-21	L7
J1-22	All Low
J1-23	L14

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Parts Activated	Points Requireing Logic High
J1-24	J1-45, J1-40, J1-39, J1-29, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-25	All Low
J1-26	J1-42, FR2
J1-27	J1-43, FR12
J1-28	All Low
J1-29	J1-45, J1-40, J1-39, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-30	J1-46, L1
J1-31	J1-47, L2
J1-32	J1-48, L21, FR9, FL9
J1-33	J1-49, FR10, FL10
J1-34	J1-6
J1-35	L15
J1-36	L16
J1-37	L11
J1-38	L17
J1-39	J1-45, J1-40, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-40	J1-45, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-41	FR1A
J1-42	J1-26, FR2
J1-43	J1-27, FL2
J1-44	FL1A
J1-45	J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-46	J1-30, L1
J1-47	J1-31, L2
J1-48	J1-32, L21, FR9, FL9
J1-49	J1-33, L20, FR110, FL10
J1-50	L5, Chassis Ground (Hole #182)
Case Ground (Hole #182)	All Low
L22	FR8, FL8
L23	FR6

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Parts Activated	Points Requiring Logic High
L24	FL6
L25	FR3
L26	FL3
L27	FR11, FL11
L28	FR7, FL7
L29	FR12
L30	FL12
L31	SL3
L32	FR4
L33	FL4
SR4	SL4
SR5	FR5
SL5	FL5

3.6 Electrical assembly (EA) (Dwg. 9311351). S&A mechanical switching shall be simulated through the use of electrical connections external to the electrical assembly as shown in Table V.

TABLE V - External connections

IDENTIFICATION	SAFE & LOCKED	SAFE & UNLOCKED	ARMED & LOCKED
S1(L&R)E1-E2	Open	Closed	Open
S1(L&R)E1-E3	Closed	Open	Closed
S2(L&R)E1-E2	Open	Open	Closed
S2(L&R)E3-E4	Closed	Closed	Open
S2(L&R)E5-E6	Open	Open	100 ohms*

*A 100 ohms = ± 10%, 1/2 watt min. resistor shall be connected between the points indicated.

3.6.1 Continuity. The electrical resistance between the pins specified in Table VI with the EA in any position shall be no greater than 1.0 ohm while applying an open electrical potential no greater than 10.0VDC.

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TABLE VI - Continuity

J1-14 to J1-7	J1-14 to J1-41
J1-14 to J1-8	J1-14 to J1-44
J1-14 to J1-12	J1-14 to J1-45
J1-14 to J1-13	J1- 6 to J1-34
J1-14 to J1-15	J1-26 to J1-42
J1-14 to J1-16	J1-27 to J1-43
J1-14 to J1-17	J1-30 to J1-46
J1-14 to J1-24	J1-31 to J1-47
J1-14 to J1-29	J1-32 to J1-48
J1-14 to J1-39	J1-33 to J1-49
J1-14 to J1-40	J1-50 to Frame Ground Pad (Hole #182)

3.6.2 Resistance. The electrical resistance shall be as specified in Table VII with the EA in the position indicated while applying an open electrical potential no greater than 10VDC.

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ined to perform the examinations and tests prescribed herein

TABLE VII - Resistance

IDENTIFICATION	J-1 CONNECTOR PINS (+) to (-)	POSITION AND REQUIREMENT (OHMS) SAFE & LOCKED	SAFE & UNLOCKED	ARMED & LOCKED
a. Condition Monitor	35 to 15	90 to 110	90 to 110	.99M to 1.1M
b. Latch Monitor, Right	3 to 15	90 to 110	.99M to 1.1M	90 to 110
c. Latch Monitor, Left	5 to 15	90 to 110	.99M to 1.1M	90 to 110
d. Fuze Fire Signal, Right	26 to 15	46 to 68	46 to 68	46 to 68
e. Fuze Fire Signal, Left	27 to 15	46 to 68	46 to 68	46 to 68
f. Rotor Monitor, Left	2 to 15	88K to 132K	88K to 132K	*NGT 2.0
g. Rotor Monitor, Right	18 to 15	88K to 132K	88K to 132K	*NGT 2.0
h. Power/Destruct to ITL Command	30 to 32	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
i. Power/Destruct to Charge Command	30 to 31	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
j. Charge Command to ITL Command	31 to 32	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
k. Charge Command to Power/Destruct	31 to 30	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
l. Ground to ITL Return	14 to 33	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm

*NGT - Not greater than

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3.6.3 Connector pins to case ground isolation resistance. The electrical resistance between J1-50 to the pins specified in Table VIII with the EA in any position shall be greater than 20 megohms while applying an open electrical potential no greater than 10.0VDC.

TABLE VIII - Isolation resistance

J1-1	J1-18	J1-34
J1-2	J1-19	J1-35
J1-3	J1-20	J1-36
J1-4	J1-21	J1-37
J1-5	J1-22	J1-38
J1-6	J1-23	J1-39
J1-7	J1-24	J1-40
J1-8	J1-25	J1-41
J1-9	J1-26	J1-42
J1-10	J1-27	J1-43
J1-11	J1-28	J1-44
J1-12	J1-29	J1-45
J1-13	J1-30	J1-46
J1-14	J1-31	J1-47
J1-15	J1-32	J1-48
J1-16	J1-33	J1-49
J1-17		

3.6.4 Five volt test. With 5.0 + 1.0VDC applied to 5V DC test input (+)J1-1 to (-)J1-15, the voltage across the pins specified in Table IX shall be as follows with the EA in the position indicated.

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TABLE IX - Five-volt test

IDENTIFICATION	CONNECTOR PINS To (-) J1-15	SAFE & LOCKED	SAFE & UNLOCKED	ARMED & LOCKED
a. Rotor Monitor, Left	J1-2	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
b. ITL TM	J1-4	4.5 ± 1.0VDC	0 ± 0.5VDC	4.5 ± 1.0VDC
c. Rotor Monitor, Right	J1-18	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
d. ARM TM	J1-19	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
e. FIRE TM	J1-20	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
f. Converter TM	J1-37	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
g. Capacitor Monitor, Left	J1-38	1.5VDC MAX	1.5VDC MAX	1.5VDC MAX
h. Capacitor Monitor, Right	J1-36	1.5VDC MAX	1.5VDC MAX	1.5VDC MAX

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3.6.5 Operating input voltage. The EA shall meet all electrical output requirements herein specified for all input voltages greater than or equal to 22.5VDC and less than or equal to 31.5VDC on the Power Destruct (+)J1-30 to (-)J1-15 or the Charge Command (+)J1-31 to (-)J1-15 or both.

3.7 Fire circuit characteristics.

3.7.1 Fire circuit charging (charge command). With the EA in the Safe and Unlocked position, the fire circuit capacitor voltage shall be 70.0VDC minimum (min.) at (+)J1-36 (Right) and (+)J1-38 (Left) to (-)J1-15 at a maximum (max.) of 2.50 seconds after 25.2 - 0.5VDC is applied to Charge Command (+)J1-31 to (-)J1-15. The initial voltage on (+)J1-36 and (+)J1-38 to (-)J1-15 shall be less than 1.5VDC prior to start of test.

3.7.1.1 Fire circuit capacitor voltage storage. With the EA in the Safe and Unlocked position, the fire circuit capacitor voltage shall remain 70.0VDC minimum (min.) at (+)J1-36 (Right) and (+)J1-38 (Left) to (-)J1-15 for six (6) seconds min., after the 25.2 - 0.5VDC charging voltage is removed from (+)J1-31 to (-)J1-15.

3.7.2 Fire circuit charging (power/destruct). With the EA in the Safe and Unlocked position, the fire circuit capacitor voltage shall be 70.0VDC minimum (min.) at (+)J1-36 (Right) and (+)J1-38 (Left) to (-)J1-15 at a maximum of 2.50 seconds after 25.2 - 0.5VDC is applied to (+)J1-30 to (-)J1-15. The initial voltage on (+)J1-36 and (+)J1-38 to (-)J1-15 shall be less than 1.5VDC prior to start of test.

3.7.3 No charging while armed. With the EA in the Armed and Locked position, the voltage on the firing circuit capacitors shall be no greater than 15.0VDC at (+)J1-36 (Right) and (+)J1-38 (Left) to (-)J1-15 after 28.0 $\pm$ 2.8VDC is applied for 6 $\pm$ 4 sec to Charge Command (+)J1-31 to (-)J1-15.

3.7.4 Fire circuit function. The left or right fire circuit is defined as having functioned at the time the Fire TM output changes from 0 $\pm$ .5V to 4.5 $\pm$ 1.0V at (+)J1-20 to (-)J1-15. The left fire circuit has functioned when the left fire circuit capacitor voltage (+)J1-38 to (-)J1-15 has discharged to less than 15.0VDC in no greater than 400msec after the Fire TM output pulse has been detected. The right fire cuircuit has functioned when the right fire circuit capacitor voltage (+)J1-36 to (-)J1-15 has discharged to less than 15.0VDC in no greater than 400msec after the Fire TM output pulse has been detected.

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3.7.4.1 Logic destruct.

3.7.4.1.1 Loss of power after arming. The fire circuits shall function in no less than 10ms and no greater than 50ms.

3.7.4.1.2 Fire upon arming after loss of power. The fire circuit shall function in no greater than 500ms prior to either Rotor Monitor (Right) (+)J1-18 to (-)J1-15 or Rotor Monitor (Left) (+)J1-2 to (-)J1-15 voltage changing from 4.5 ± 1.0 to 0 ± 0.5VDC.

3.7.4.1.3 Power destruct threshold. The fire circuit shall function when the voltage applied to Power/Destruct (+)J1-30 to (-)J1-15 is between 18.0 and 21.0VDC.

3.7.4.2 Fuze fire function.

3.7.4.2.1 No fire signal A. Neither fire circuit shall function when a 5.0 ± 0.25V pulse for a duration of 200 ± 10 microseconds with a rise time no greater than one (1) microsecond is applied to (+)J1-27 (Left) and (+)J1-26 (Right) to J1-15 fuze signal input. At completion of the No Fire signal both fire circuit capacitor monitors shall be 70VDC min.

3.7.4.2.2 No fire signal B. Neither fire circuit shall function when a 10.0 ± 0.5V pulse for a duration of 2.0 + 0.5 microseconds with a rise time no greater than one (1) microsecond is applied to (+)J1-27 (Left) and (+)J1-26 (Right) to (-)J1-15 fuze signal input. At completion of the No Fire signal both fire circuit capacitor monitors shall be 70VDC min.

3.7.4.2.3 No fire signal C. Neither fire circuit shall function when a 30.0 ± 1.5V pulse for a duration of 0.4 ± 0.05 microseconds with a rise time no greater than 0.08 microseconds is applied to (+)J1-27 (Left) and (+)J1-26 (Right) to (-)J1-15 fuze signal input. At completion of the No Fire signal both fire circuit capacitor monitors shall be 70VDC min.

3.7.4.2.4 Fuze fire signal. Both fire circuits shall function when a fuze fire signal of 18.0 - 2.0V for a duration of 90 - 10 microseconds with a rise time NGT 20 microseconds is applied to (+)J1-27 (Left) and (+)J1-26 (Right) to (-)J1-15 .

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3.8 EA interface outputs (J-1 connector).

3.8.1 Power TM threshold. The Power TM monitor at (+)J1-21 to (-)J1-15 shall be 4.5 ± 1.0VDC when the Power/Destruct input at (+)J1-30 to (-)J1-15 is greater than 25.0VDC. The Power TM monitor shall be 0 ± 0.5 when the Power/Destruct input is less than 21.0VDC. This transition shall occur at 23.0 ± 2.0VDC.

3.8.2 Voltage outputs. With the EA in the position indicated in Table X the voltage requirements shall be as specified when the input voltage as indicated is applied to Power/Destruct (+)J1-30 to (-)J1-15 for ten (10) seconds min. prior to measurements.

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TABLE X - Interface outputs (J1)

IDENTIFICATION	CONNECTOR PINS To (-) J1-15	INPUT VOLTAGE POWER/DESTRUCT (23.0 - 0.5VDC) SAFE & LOCKED	(23.0 - 0.5VDC) SAFE & UNLOCKED	(31.0 + 0.5VDC) ARMED & LOCKED
a. Rotor Monitor, Left	J1-2	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
b. ITL TM	J1-4	4.5 ± 1.0VDC	0 ± 0.5VDC	4.5 ± 1.0VDC
c. Rotor Monitor, Right	J1-18	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
d. ARM TM	J1-19	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
e. FIRE TM	J1-20	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
f. 5VDC Monitor	J1-23	4.5 ± 1.0VDC	4.5 ± 1.0VDC	4.5 ± 1.0VDC
g. 100V Capacitor Monitor (Right)	J1-36	70 to 110VDC	70 to 110VDC	80 to 110VDC
h. Converter TM	J1-37	4.5 ± 1.0VDC	4.5 ± 1.0VDC	4.5 ± 1.0VDC
i. 100V Capacitor Monitor (Left)	J1-38	70 to 110VDC	70 to 110VDC	80 to 110VDC

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3.9 Environmental and mechanical.

3.9.1 Temperature cycling. Semiconductors, when tested as specified in MIL-STD-750, Method 1051.1 and MIL-STD-202, Method 107, Test Condition F-1, shall meet the final electrical requirements after exposure.

3.9.2 Constant acceleration. Semiconductors, when tested in the Y1 and Z1 axes in accordance with MIL-STD-750, Method 2006, at 20000g's, shall meet the final electrical requirements.

3.9.3 Seal.

3.9.3.1 Fluorocarbon gross leak. Semiconductors shall meet the requirements specified in MIL-STD-750, Method 1071, Test Condition C.

3.9.3.2 Fine leak. Semiconductors shall meet the requirements specified in MIL-STD-750, Method 1071, Test Condition H or G.

3.9.4 Burn-in. Semiconductors shall be subjected to burn-in testing in accordance with the details herein and MIL-STD-750, Method 1038 or 1039, Test Condition B, for 168 hours at 125°C minimum, and shall meet the final electrical requirements after exposure.

3.9.5 Thermal shock. Semiconductors, when tested as specified in MIL-STD-750, Method 1056, Test Condition B, 15 cycles, shall meet the final electrical requirements after exposure.

3.9.6 Mechanical shock. Semiconductors, when subjected to the shock tests specified in MIL-STD-750, Method 2016, 1500g's, 0.5 milliseconds, 6 shock pulses X1, Y1, and Z1 axes, shall meet the final electrical requirements after exposure.

3.9.7 Life cycle. Transformers and inductors, when subjected to the life cycle tests specified in MIL-T-27, 168 hours at 125°C shall meet the final electrical requirements after exposure.

3.10 Marking.

3.10.1 Lot identification code. Components specified in Table XI shall be marked by a code indicating the week inspection including burn-in was completed for the lot accumulation period. The marking shall be as specified in MIL-STD-1285.

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TABLE XI - Lot identification code

DRAWING	Part Description
9311342	Capacitor, 86 microfarad
9311353	Choke
9311254	Transformer
9311342	SCR, 2N 2326 JANTX
9311356	Bridge Rectifier
9311359	Transistor, 2N6193
9311372	Capacitor, 68 Microfarad

3.10.2 Test data. Test data from tests specified in 4.4.2.4.1 and 4.4.2.15.1 shall be submitted if requested by the procuring activity. Variable data shall be submitted for those requirements specified in 3.3.2, 3.3.3.1, 3.3.3.2, 3.3.3.4, 3.3.4, 3.4.1, 3.4.2.1 thru 3.4.2.5, 3.4.2.6.1, 3.4.2.6.2, 3.4.2.7, 3.4.3.1, 3.4.3.2, 3.4.3.3, 3.4.3.4, 3.4.4.1, 3.6.2, 3.6.4, 3.7.1, 3.7.1.1, 3.7.2, 3.7.3, 3.7.4, 3.7.4.1.1, 3.7.4.1.2, 3.7.4.1.3, 3.8.1 and 3.8.2. Attribute data (go, no-go) may be submitted for all other requirements.

3.11 Workmanship. All parts and assemblies shall be fabricated in a careful and workmanlike manner. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Drawings	Part Description	Quantity
9311342	Capacitor, 86 microfarad	5
9311342	SCR, JANTX 2N2326	5
9311342	Semiconductor JANTX IN485B	5
9311342	Fire Board Assembly	5
9311345	Fire Circuit Board	5
9311346	Insulator	5
9311353	Choke	5
9311354	Transformer	5
9311356	Bridge Rectifier	5
9311359	Transistor 2N6193	5
9311360	Connector	5
9311361	Flexible Cable	5
9311371	Logic Circuit Board	5
9311371	Logic Circuit Board Assembly	5
9311372	Capacitor, 68 microfarad	5
9321558	Connector Cable Assy, prior to potting	5
9321558	Connector Cable Assembly	5
9321567	Mounting Plate (rear)	5
9321568	Bracket	5
9321569	Mounting Plate Assy	5
9321570	Connector Assy	5

4.3.1.1 Homogeneity. The first article shall comply with the homogeneity criteria of MIL-A-48078, Paragraph 4.4.1.

4.3.2 Inspections to be performed. First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the government to any or all the requirements of the applicable drawings and to the inspections of Table XII. All inspections listed in Table XII shall be classified Major unless otherwise specified.

4.3.3 Rejection. See MIL-A-48078.

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TABLE XII - First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63335C

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See 4.3.1 NEXT HIGHER ASSEMBLY
4.3.4	Electrical Parts and Assemblies; XM143				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Assemblies, Subassemblies and Components (Drawing list - See 4.3.1)	See 4.3.1		3	4.4.2.1 thru 4.4.2.23
	Examination for Defects				
	Additional Tests to be performed				
	Fire Board Assembly (Dwg. 9311342)	See 4.3.1		3.4	Test
	Bridge Rectifier (Dwg. 9311343)	See 4.3.1		3.10	4.4.2.9.1
	Transistor High Power, Silicon P.N.P. (2N6193) (Dwg. 9311342)	See 4.3.1		3.10	4.4.2.10.1

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Tests. Paragraphs, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Level II.

4.4.3 Testing. N/A

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311342
4.4.2.1	Capacitor (86 Microfarad)				NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	Capacitance Bridge/ MIL-C-39006/22-0200
102	Dissipation factor		0.40%	3.2	Capacitance Bridge/ MIL-C-39006/22-0200
103	Marking missing or incorrect		0.40%	3.7	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.2	Semiconductor, Device Silicon Diode JANTX IN485B		1 1		9311342/MIL-S-19500
					NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward voltage, V_F		0.40%	3.2	MIL-STD-750/4011 or Teradyne
102	Reverse current, I_R		0.40%	3.2	MIL-STD-750/4016 or Teradyne
103	Band missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Rectifier, Controlled, Silicon JANTX2N2326	1	1	9311342/MIL-S-19500
				NEXT HIGHER ASSEMBLY: 9311342

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Reverse blocking current, I_{RXM}		0.40%	3.2	MIL-STD-750/4421 or Teradyne
102	Forward blocking current, I_{FXM}		0.40%	3.2	MIL-STD-750/4206 or Teradyne
103	Gate trigger voltage, V_{GT}		0.40%	3.2	MIL-STD-750/4221 or Teradyne
104	Gate trigger current I_{GT}		0.40%	3.2	MIL-STD-750/4221 or Teradyne
105	Forward on voltage V_{FM}		0.40%	3.2	Teradyne
106	Marking		0.40%	3.7	Visual
Minor					
201	Evidence of poor workmanship		0.40%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311342
4.4.2.4	Fire Board Assembly (Visual)				NEXT HIGHER ASSEMBLY 9311357
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder in accordance with MIL-STD-1460		0.40%	3.2	Visual
102	Capacitor or diode polarity incorrect		0.40%	3.2	Visual
103	Incorrect component, damaged or missing component		0.40%	3.2	Visual
104	Pins on wrong side of board		0.40%	3.2	Visual
105	Maximum height of component leads and soldering		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311342
4.4.2.4.1	Fire Board Assembly (Electrical Test)				NEXT HIGHER ASSEMBLY 9311357
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Continuity		100%	3.3.1	Test
102	Resistance		100%	3.3.2	Test
103	No charging		100%	3.3.3.3	Test
104	Fire circuit charging		100%	3.3.3.1	Test
105	Fire circuirt capacitor storage		100%	3.3.3.2	Test
106	Fuze Fire Signal		100%	3.3.4.5	Test
107	No Fire Signal "A"		100%	3.3.4.2	Test
108	No Fire Signal "B"		100%	3.3.4.3	Test
109	No Fire Signal "C"		100%	3.3.4.4	Test
110	Logic destruct signal		100%	3.3.4.1	Test
111	Fire circuit stabilized voltage		100%	3.3.3.4	Test

NOTES:

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QUALITY CONFORMANCE INSECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Fire Circuit Board		9311345
			NEXT HIGHER ASSEMBLY 9311342

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole locations		0.40%	3.2	Gage
102	Hole diameters		0.40%	3.2	Gage
103	Board cracked, chipped, warped or twisted		0.40%	3.2	Visual
104	Conductive paths reduced by 1/3 or more of the width		0.40%	3.2	Visual
105	Conductor not bonded to board or peeling		0.40%	3.2	Visual
106	Conductor areas not plated		0.40%	3.2	Visual
107	Bridging of conductors		0.40%	3.2	Visual
108	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

AF Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311346
4.4.2.6	Insulator				NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness	10/ sack		3.2	Micrometer
102	Improper configuration, color or part	10/ sack		3.2	Visual
103	Evidence of flash in holes or on surface of feet	20/ sack		3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311353
4.4.2.7	Choke (Visual)				NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder in accordance with MIL-STD-1460 (4 places)		0.40%	3.2	Visual
102	Molding compound cure improper		0.40%	3.2	Visual/Manual
103	Evidence of excessive adhesive		0.40%	3.2	Visual
104	Molding compound above flush		0.40%	3.2	Visual
105	Adhesive/molding compound shelf life expired		0.40%	3.2	Visual
106	Marking (after final electrical)		0.40%	3.7	Visual
107	Cup height, max.		0.40%	3.2	Gage
108	Cup diameter, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311353
4.4.2.7.1	Choke (Electrical Test)				NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Life cycles with 60 milliamps at temp		0.40%	3.9.7	Test/MIL-T-27
102	Inductance		0.40%	3.2	Test
103	Resistance		0.40%	3.2	Test

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9311354
4.4.2.8	Transformer (Visual)		NEXT HIGHER ASSEMBLY 9311342

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder in accordance with MIL-STD-1460 (8 places)		0.40%	3.2	Visual
102	Molding compound cure improper 1/		0.40%	3.2	Visual/Manual
103	Evidence of excessive adhesive		0.40%	3.2	Visual
104	Molding compound above flush		0.40%	3.2	Visual
105	Adhesive/molding compound shelf life expired		0.40%	3.2	Visual
106	Marking (after final electrical)		0.40%	3.10	Visual
107	Cup height, max.		0.40%	3.2	Gage
108	Cup diameter, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:
1/ Tackiness, cracks, lack of adhesion or excessive voids shall be evidence of improper cure.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311354
4.4.2.8.1	Transformer (Electrical Test)				NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Life cycle with 48VDC		0.40%	3.9.7	Test/MIL-T-27/ Dwg. 9311354
102	Dynamic electrical characteristic		0.40%	3.2	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311356
4.4.2.9	Bridge Rectifier				NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Burn-in		100%	3.9.4	Test
102	Forward voltage drop per leg		100%	3.2	Test
103	Leakage current per leg		100%	3.2	Test
104	Marking (after final electrical)		0.40%	3.10	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.9.1	Bridge Rectifier (Environmental, Destructive Test)		1 1	9311356
				NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Sequence of Environments Required				
Major					
101	Temperature cycling		0.40%	3.9.1	Test
102	Constant acceleration		0.40%	3.9.2	Test
103	Thermal shock		0.40%	3.9.5	Test
104	Mechanical shock		0.40%	3.9.6	Test
105	Forward voltage drop per leg		0.40%	3.2	Test
106	Leakage current per leg		0.40%	3.2	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.10	Transistor, High Power, Silicon P.N.P. (2N6193)		1 2		9311359
					NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Burn-in		100%	3.9.4	Test
102	Collector cut-off current with emitter open		100%	3.2	Test
103	Emitter-cut-off current		100%	3.2	Test
104	Collector-to-emitter sustaining voltage with base open		100%	3.2	Test
105	Collector-to-emitter saturation voltage		100%	3.2	Test
106	Base-to-emitter voltage		100%	3.2	Test
107	DC forward current transfer ratio		100%	3.2	Test
108	Marking (after final electrical)		0.40%	3.10	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

AMSC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.10.1	Transistor, High Power, Silicon P.N.P. (2N6193)			2 2	9311359
					NEXT HIGHER ASSEMBLY 9311342
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Sequence of Environmentals Required				
Major					
101	Temperature cycling		0.40%	3.9.1	Test
102	Constant acceleration		0.40%	3.9.2	Test
103	Seal		0.40%	3.9.3	Test
104	Thermal shock		0.40%	3.9.5	TEst
105	Mechanical shock		0.40%	3.9.6	Test
106	Electrical		0.40%	3.2	Test/4.4.2.10

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311360
4.4.2.11	Connector				NEXT HIGHER ASSEMBLY 9321570
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of threaded post		0.40%	3.2	GO/NO-GO Gage
102	Pins damaged or missing		0.40%	3.2	Visual
103	Protective caps missing		0.40%	3.2	Visual
104	Plating missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311361
4.4.2.12	Flexible Cable				NEXT HIGHER ASSEMBLY 9321558
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Plating on pads missing		0.40%	3.2	Visual
102	Breaks in circuitry		0.40%	3.2	Visual
103	Delamination		0.40%	3.2	Visual
104	Tears or nicks in outside edge		0.40%	3.2	Visual
105	Location of holes		0.40%	3.2	Visual
106	Solder Wicking		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9311371
4.4.2.13	Logic Circuit Board				NEXT HIGHER ASSEMBLY 9311372
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole location		0.40%	3.2	Gage
102	Hole diameter		0.40%	3.2	Gage
103	Board cracked, chipped, warped or twisted		0.40%	3.2	Visual
104	Conductive paths defects reduced by 1/3 or more of the width		0.40%	3.2	Visual
105	Conductor not bonded to board or peeling		0.40%	3.2	Visual
106	Conductor areas not plated		0.40%	3.2	Visual
107	Bridging of conductors		0.40%	3.2	Visual
108	Annular ring encirclement improper		0.40%	3.2	Visual
109	Plated through hole defects - more than three (3) plating voids per hole		0.40%	3.2	Visual 1/

NOTES:
1/ A sample of three (3) holes of each diameter on each board shall be inspected.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9311371
4.4.2.13	Logic Circuit Board				NEXT HIGHER ASSEMBLY 9311372
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110	Plated through hole defects - voids in excess of 10% of surface of hole		0.40%	3.2	Visual
111	Plated through hole defects - voids at interface of hole		0.40%	3.2	Visual
112	Plated through hole defects - separation around circumference		0.40%	3.2	Visual
113	Plating through hole thickness		0.40%	3.2	Test 2/ - Caviderm Unit Process Assy, Inc.
114	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:
2/ A sample of three (3) holes of each diameter on each board shall be inspected for a 1.2 caviderm reading (required for 1 Mil in the hole thickness).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311372
4.4.2.14	Capacitor (68 Microfarad)				NEXT HIGHER ASSEMBLY 9311372
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	Capacitance Bridge/ MIL-C-39006/22-0110
102	Dissipation factor		0.40%	3.2	Capacitance Bridge/ MIL-C-39006/22-0110
103	Marking missing or incorrect		0.40%	3.10	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.15	Logic Circuit Board Assembly (Visual)		1 OF 1		9311372
					NEXT HIGHER ASSEMBLY 9311357
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder in accordance with MIL-STD-1460		0.40%	3.2	Visual
102	Capacitor polarity incorrect		0.40%	3.2	Visual
103	Incorrect component, damaged or missing component		0.40%	3.2	Visual
104	Pins on wrong side of board		0.40%	3.2	Visual
105	Maximum height of component leads and soldering		0.40%	3.2	Gage
106	Height of hybrid off board		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15.1	Logic Circuit Board Assembly (Electrical Test)	1 OF 2	9311372
			NEXT HIGHER ASSEMBLY 9311357

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Continuity and resistance		100%	3.4.1	Test
102	TM monitors (with charge command and right convertor input)		100%	3.4.2.1	Test
103	Fire TM		100%	3.4.2.2	Test
104	Arm TM		100%	3.4.2.3	Test
105	ITL TM		100%	3.4.2.4	Test
106	Convertor TM		100%	3.4.2.5	Test
107	Capacitor storage		100%	3.4.2.7	Test
108	Destruct pulse, Power Up (with Charge Command input)		100%	3.4.2.6.1	Test
109	Destruct pulse, convertor input removed Charge Command input		100%	3.4.2.6.2	Test
110	TM monitors (with Power/Destruct and left convertor input)		100%	3.4.3.1	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.15.1	Logic Circuit Board Assembly (Electrical Test)		2 2		9311372
					NEXT HIGHER ASSEMBLY 9311357
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Power TM threshold (with Power/ Destruct input)		100%	3.4.3.2	Test
112	Destruct pulse threshold (with Power/ Destruct input)		100%	3.4.3.3	Test
113	Convertor TM threshold (with Power/ Destruct input)		100%	3.4.3.4	Test
114	Five volt test, Destruct pulse		100%	3.4.4.1	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.16	Molding Compound		9311390
			NEXT HIGHER ASSEMBLY 9311353/9321561

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect material		As req'd	3.2	Visual
102	Shelf life expired		As req'd	3.2	Visual

NOTES:

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Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.17	Connector Cable Assembly, Prior to Potting		1 1		9321558
					NEXT HIGHER ASSEMBLY 9311357
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering as specified in MIL-STD-1460		0.40%	3.2	Visual
102	Maximum height of pins and cable after soldering		0.40%	3.2	Gage
103	Continuity and isolation		100%	3.5.1	Test
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Connector Cable Assembly, After Potting		9321558
			NEXT HIGHER ASSEMBLY 9311357

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pins above potting surface		0.40%	3.2	Visual
102	Holes or cavities in potting compound surface only		0.40%	3.2	Visual
103	Evidence of entrapped air		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Mounting Plate (Rear)	1 OF 1	9321567
			NEXT HIGHER ASSEMBLY: 9321569

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of holes		0.40%	3.2	Gage
102	Pitch diameter and depth		0.40%	3.2	Gage
103	Length and width of cutout		0.40%	3.2	Gage
104	Length and width of cutout for connector		0.40%	3.2	Gage
105	Depth of cutout for connector rear surface		0.40%	3.2	Gage
106	Length and width of outer edge cutout for cover surface		0.40%	3.2	Gage
107	Depth of outer edge cutouut for cover surface		0.40%	3.2	Gage
108	Location of drive pin holes		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9321568
4.4.2.20	Bracket				NEXT HIGHER ASSEMBLY 9321568
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mounting hole diameter, min.		0.40%	3.2	Gage
102	Location of holes		0.40%	3.2	Gage
103	Evidence of sharp edges		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

AMSC Form 1570, 1 Feb 85 Replac DRSMC-QA (D) Form 160, 1 Aug 83, which may not be us

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9321569
4.4.2.21	Mounting Plate Assembly				NEXT HIGHER ASSEMBLY 9321570
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Brackets loose		0.40%	3.2	Visual
102	Drive pins not seated		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

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MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9321570
4.4.2.22	Connector Assembly				NEXT HIGHER ASSEMBLY 9321558
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Connector flange above flush		0.40%	3.2	Visual/Manual
102	Excess adhesive		0.40%	3.2	Visual
103	Evidence of adhesive on pins (2 sides)		0.40%	3.2	Visual
104	Orientation of connector outer shell in mounting plate assy.		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.23	Electrical Assembly (EA) Visual Prior to Conformal Coating and Adhesive		1 1		9311357
					NEXT HIGHER ASSEMBLY 9311101
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder in accordance with MIL-STD-1460		0.40%	3.2	Visual/MIL-STD-1460
102	Pin not soldered		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311357
4.4.2.24	Electrical Assembly (Visual After Conformal Coating)				NEXT HIGHER ASSEMBLY 931101
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Conformal coating not continuous		0.40%	3.2	Visual
102	Evidence of excess conformal coating		0.40%	3.2	Visual
103	Conformal coating around mounting holes		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

AMSC Form 1570, 1 Feb 85 Repla DRSMC-QA (D) Form 160, 1 Aug 83, which may not be u

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311357
4.4.2.25	Electrical Assembly Electrical Operation				NEXT HIGHER ASSEMBLY 9311101
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Continuity	100%	1/	3.6.1	Test
102	Resistance (Safe & Locked)	100%		3.6.2	Test
103	Connector pins to case ground isolation resistance	100%		3.6.3	Test
104	Fire circuit charging (Charge Command)	100%		3.7.1	Test
105	Fire circuit capacitor voltage storage	100%		3.7.1.1	Test
106	Fuze Fire function	100%		3.7.4.2	4.5.4/Test
107	No charging while armed	100%		3.7.3	Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63335C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311357
4.4.2.26	Electrical Assembly Electrical Operation				NEXT HIGHER ASSEMBLY 9311101
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
108	Fire circuit charging (Power/Destruct)	100%	1/	3.7.2	Test
109	Fire upon arming after loss of power	100%		3.7.4.1	4.5.2/Test

NOTES:
1/ Accept on 0 defects, reject on 1 or more defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER 9311357	
4.4.2.27	Electrical Assembly Electrical Operation			NEXT HIGHER ASSEMBLY 9311101	
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
110	Voltage Output (Safe and Locked)	100%		3.8.2	Test
111	Voltage Output (Safe and Unlocked)	100%		3.8.2	Test
112	Voltage Output (Armed and Locked)	100%		3.8.2	Test
113	Loss of Power After Arming	100%		3.7.4.1.1	Test/4.5.1
114	Five-Volt Test (Safe and Locked)	100%		3.6.4	Test
115	Resistance (Safe and Unlocked	100%		3.6.2	Test
116	Five Volt Test (Safe and Unlocked	100%		3.6.4	Test
117	Five-Volt Test (Armed and Locked)	100%		3.6.4	Test
118	Resistance (Armed and Locked)	100%		3.6.2	Test
119	Power TM Threshold	100%		3.8.1	4.5.5/Test
120	Power/Destruct Threshold	100%		3.7.4.1.3	4.5.3/Test

NOTES:
1/ Accept on 0 defects, reject on 1 or more defects.

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4.5 Methods of inspections.

4.5.1 Procedure for loss of power after arming test.

4.5.1.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. Electrical Assembly Armed and Locked.

b. 31.0 + 0.5VDC applied to Power/Destruct.

c. Left capacitor monitor @ 80.0VDC min.

d. Right capacitor monitor @ 80.0VDC min.

4.5.1.2 Dynamic conditions.

a. Remove 31.0 + 0.5VDC on Power/Destruct.

b. Measure time (3.8.4.1.1) to function.

4.5.2 Procedure for fire upon arming after loss of power test.

4.5.2.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

a. Electrical Assembly Safe and Unlocked.

b. 25.2 - 0.5VDC on Power/Destruct.

c. Left capacitor monitor @ 70.0VDC min.

d. Right capacitor monitor @ 70.0VDC min.

4.5.2.2 Dynamic conditions.

a. Remove 25.2 - 0.5VDC on Power/Destruct.

b. Electrical Assembly Armed and Locked.

c. Measure time (3.8.4.1.2) to function.

4.5.3 Procedure for power/destruct threshold test.

4.5.3.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

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a. Electrical Assembly Armed and Locked.

b. 23.0 + 2.0VDC applied to Power/Destruct.

c. Left capacitor monitor @ 70VDC min.

d. Right Capacitor Monitor @ 70VDC min.

4.5.3.2 Dynamic conditions. Decrease Power/Destruct input voltage ((+) J1-30 to (-) J1-15) with a voltage step no greater than 25 millivolt (mv) and record input voltage level at the time the fire circuit functions (see 3.7.4).

4.5.4 Procedure for fuze fire function.

4.5.4.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

a. Electrical Assembly Armed and Locked.

b. 23.0 - 0.5VDC applied to Power/Destruct.

c. Left capacitor monitor @ 70.0VDC min.

d. Right capacitor monitor @ 70.0VDC min.

4.5.4.2 Dynamic conditions.

a. Apply NO Fire Signal A
b. Apply NO Fire Signal B
c. Apply NO Fire Signal C
d. Apply Fuze Fire Signal

4.5.5 Procedure for power TM threshold test.

4.5.5.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. Electrical Assembly Armed and Locked.

b. 28.0 $\pm$ 3.0VDC applied to Power/Destruct.

c. Left capacitor monitor @ 70.0VDC min.

d. Right capacitor monitor @ 70.0VDC min.

4.5.5.2 Dynamic conditions. Decrease Power/Destruct input voltage using a voltage step no greater than 25 multivolts (mv) and record input voltage at the time the Power TM Monitor changes states (see 3.7.1).

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5. PACKAGING - N/A

6. NOTES

6.1. Intended use - This specification is intended to be used in conjunction with MIL-S-63257, Safety and Arming Device, GM: M143 Loading, Assembling and Packing For. In the event of conflict, MIL-S-63257 shall take precedence.

6.2 Ordering data - See MIL-A-48078.

6.2.1 Data requirements - Deliverable data required by this specification is cited in the following paragraphs

Paragraph	Data Requirement	Applicable DID
3.10	Test and Demonstration Reports	DI-T-1906

(Copies of data item description required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 Submission of designs for approval - See 6.2.3 to MIL-A-48078. Submit equipment designs, as required, to ATTN: AMSMC-QAR-I, Dover, New Jersey 07801-5001.

6.4 Drawings - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is new under the cognizance of ARDC.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing activity:
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(Project: 1336-A500)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-63335C	S&A DEVICE, GM: M143 ELECTRICAL PARTS AND ASSEMBLIES FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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